VRL Advanced AI Systems — Technical Documentation Pack

Prepared for: Netcare AI Committee

Vision Research Labs | April 2, 2026

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1. Executive Summary

VRL Advanced AI Systems is a suite of five production-deployed healthcare AI products purpose-built for the South African private healthcare market. Each product addresses a specific workflow pain point — from clinical documentation to claims processing to patient routing.

Key differentiators:

- Trained on SA-specific data: ICD-10 (WHO, not US ICD-10-CM), CCSA tariff codes, NAPPI medicine codes, SA medical scheme rules
- Designed to **complement existing infrastructure** (CareOn, Healthbridge, MediKredit, SwitchOn) — not replace it
- All products are live and production-deployed on Vercel's global edge network

- POPIA-aware architecture with audit logging, consent tracking, and data sovereignty (EU-West/SA-proximate hosting)

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2. Product Suite Overview

2.1 Doctor OS — AI Clinical Copilot

- **URL**: https://doctor-os.vercel.app
- **Purpose**: AI-powered clinical assistant with real-time medical scribe, ICD-10 coding, and full practice management
- **Scale**: 15 pages, 10 API routes, 38 AI tools, 10 clinical engines, 18 database models
- **Key capabilities**:
  - Real-time audio transcription (Gemini 2.5 Flash) with clinical note generation
  - Automatic ICD-10 code assignment from consultation transcripts
  - CCSA tariff code suggestion
  - Drug interaction checking (NAPPI database)
  - Automated SOAP note generation
  - Patient intake, vitals capture, billing

2.2 VisioCode — AI Clinical Coding System

- **URL**: https://visiocode.vercel.app
- **Purpose**: Standalone clinical coding tool for medical practices and billing departments
- **Scale**: 18 pages, 24 API routes, 18 AI agent tools (12 core + 6 senior coder)
- **Key capabilities**:
  - ICD-10 code search and validation (41,008 SA MIT codes with PMB/CDL/gender/age flags)
  - NAPPI medicine code lookup (482,695 active records)
  - CCSA tariff code assignment (10,304 codes with GEMS 2026 Rand rates)
  - AI-assisted code suggestion from clinical notes (Gemini 2.5 Flash agent)
  - Batch coding for claims submissions
  - 250+ code pair violation rules, unbundling detection
  - HMAC-SHA256 claim signing/verification
  - 15-signal fraud scoring (Fraud Sentinel)
  - PMB/CDL explorer (93 DTPs + 27 chronic conditions)
  - Rate limiting via Upstash Redis (60 req/min general, 10 AI/min)

2.3 Netcare Health OS — Enterprise Healthcare Platform

- **URL**: https://healthos.visiocorp.co
- **Purpose**: Full enterprise healthcare operating system with 6 integrated AI products

- **Scale**: 181 pages, 244 API routes, 56 database models, 60,000+ lines of code
- **White-label multi-tenant**: Per-practice branding, subdomain, feature flags per tenant
- **7 Vercel cron jobs**: reminders (hourly), claims retention (monthly), reports (daily), ML reinforcement (daily), registry update (weekly), engagement (15min), email polling (2hr)
- **6 AI Products**:
  1. **CareOn Bridge** — HL7v2 parser → FHIR mapper → AI advisory engine for Netcare's CareOn EMR
  2. **FHIR Hub** — Full FHIR R4 server with SMART configuration, resource CRUD, validation
  3. **Switching Engine** — 3-vendor claims switching (Healthbridge, SwitchOn, MediKredit), EDIFACT MEDCLM, batch/preauth/ERA
  4. **Healthbridge Claims** — AI autofill, rejection prediction, batch processing, remittance reconciliation, eligibility checking
  5. **Claims Analyzer** — Pattern detection, rule engine, geo-fraud detection, auto-correction, scoring
  6. **WhatsApp Module** — Twilio-based broadcast, incoming routing, clinic finder with geo-search

2.4 Claims Rejection Analyzer

- **URL**: https://claims-rejection-analyzer.vercel.app
- **Purpose**: AI-powered pre-submission validation + rejection analysis for medical scheme claims
- **Scale**: 10 authenticated, rate-limited API routes
- **Key capabilities**:
  - Pre-submission ICD-10-ZA validation engine (427 ICD-10 codes, 276 NHRPL tariff codes, 79 NAPPI codes)
  - 7 scheme profiles (Discovery, GEMS, Bonitas, Medshield, Momentum, Bestmed)
  - 13 core validation rules + scheme-specific + tariff cross-validation
  - Auto-correction engine for high-confidence deterministic fixes
  - Pattern learning from historical rejections
  - Healthbridge EDI auto-detection (parse switch export format)
  - AI models: Gemini (primary, cost-optimized) + Claude (fallback)
  - POPIA compliant: PII anonymized, consent gate, 12-month retention, audit trail
  - Raw PDF report generation (zero dependencies)
- **Financial impact**: Average SA rejection rate 15-20% (R800/claim). Recovery rate with pre-validation: 85%. Monthly savings per practice: R51K-68K

2.5 Patient Flow AI

- **URL**: https://patient-flow-ai.vercel.app
- **Purpose**: Intelligent patient routing, queue management, and no-show prediction
- **Scale**: 13 pages, 19 database tables, FlowBot AI agent (45 tools), Supabase shared DB

- **Key capabilities**:
  - No-show prediction: dual-model (Gemini 3 Flash AI + statistical logistic regression), 90% accuracy on first retrain, ~R0.002/prediction
  - Capacity forecasting with revenue impact analysis
  - Patient flow Kanban board (waiting → in consultation → done → no-show + blockers)
  - Doctor consultation pattern analysis
  - Automated patient engagement sequences (multi-step journeys)
  - Health campaigns (recall, vaccination, screening)
  - Email triage with AI categorization
  - WhatsApp/SMS integration via Twilio (inbound webhook, broadcast up to 50, button menus, emergency call routing)
  - CareOn hospital HL7 bridge (ADT, lab results, orders)
  - Medical aid verification via Eligibility API
  - Morning briefing: complete picture (bookings, predictions, overdue recalls, tasks)
  - Self-learning: records outcomes → weekly retrain → adjusts weights

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3. Architecture & Tech Stack

3.1 Frontend

Component	Technology	Version
Framework	Next.js (App Router)	16.1.6 — 16.2.1
Language	TypeScript	5.x (strict mode)
UI Components	shadcn/ui + Radix UI	Latest
Styling	Tailwind CSS	4.x
Animations	Framer Motion	12.x
Icons	Lucide React	Latest

3.2 Backend

Component	Technology	Details
Runtime	Node.js	Vercel Serverless + Edge Functions
API	Next.js API Routes	RESTful, per-route rate limited
ORM	Prisma	7.4 — 7.6 (adapter-based)
Database	PostgreSQL (Supabase)	Managed, EU-West-1 region

Component	Technology	Details
Auth	Supabase Auth	Cookie-based SSR, Row Level Security
Rate Limiting	Upstash Redis	Per-route sliding window
Email	Resend	Transactional + notifications
Analytics	Vercel Analytics	Performance + usage tracking

3.3 AI Stack

Component	Technology	Use Case
Primary LLM	Claude Sonnet 4.6 (Anthropic)	Clinical reasoning, tool calling, agent loops
Reasoning LLM	Claude Opus 4.6 (Anthropic)	Complex clinical decisions
Fast LLM	Claude Haiku 4.5 (Anthropic)	Quick lookups, classification
Transcription	Gemini 2.5 Flash (Google)	Real-time audio → clinical text
Image/Multimodal	Gemini 2.5 Pro (Google)	Document analysis, image interpretation
Voice	ElevenLabs	Text-to-speech for agent responses
AI Framework	Vercel AI SDK	v6.x — unified provider interface
Agent Pattern	Tool-calling agent loop	`streamText` + `stopWhen: stepCountIs(N)`

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4. AI Models & Agent Architecture

4.1 Agent Design Pattern

All VRL products use a **tool-calling agent architecture**:

```
User Query → LLM (Claude) → Tool Selection → Tool Execution → Result → LLM →
Response
                    ↑                                                       |
                    └──────────────── Loop (up to N steps) ─────────────────┘
```

- Agents are given a set of **typed tools** (functions with Zod-validated inputs/outputs)
- The LLM autonomously selects which tools to call based on user intent
- Multi-step reasoning: the agent can chain multiple tools in sequence
- All tool calls are logged for audit/compliance

4.2 Doctor OS Agent (38 Tools)

The Doctor OS agent has 38 tools organized into clinical workflow categories:

Category	Tools	Examples
Patient Management	~8	Create patient, search, update demographics, merge records
Clinical Documentation	~6	Start consultation, transcribe audio, generate SOAP notes
Clinical Coding	~5	Search ICD-10, suggest codes, validate code combinations
Prescriptions	~4	Search NAPPI, check interactions, generate scripts
Billing & Claims	~5	Create invoice, assign tariff codes, submit claims
Practice Management	~5	Schedule, queue management, referrals
Analytics	~3	Revenue reports, patient stats, coding accuracy
System	~2	Memory, context management

4.3 Practice-Scoped Data Access

All tools use a `getPracticeId()` closure pattern ensuring data isolation:

- Every database query is scoped to the authenticated practice
- No cross-practice data leakage possible at the ORM level
- Supabase Row Level Security provides a second enforcement layer

4.4 Knowledge-Grounded Responses

AI agents are grounded in a **300MB+ health knowledge base** (see Section 9) to prevent hallucination on clinical content. The AI does not generate medical codes from general knowledge — it searches verified databases.

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5. Data Storage & Flow

5.1 Database Architecture

Primary Database: PostgreSQL hosted on Supabase (managed, EU-West-1)

Netcare Health OS Models (56 tables):

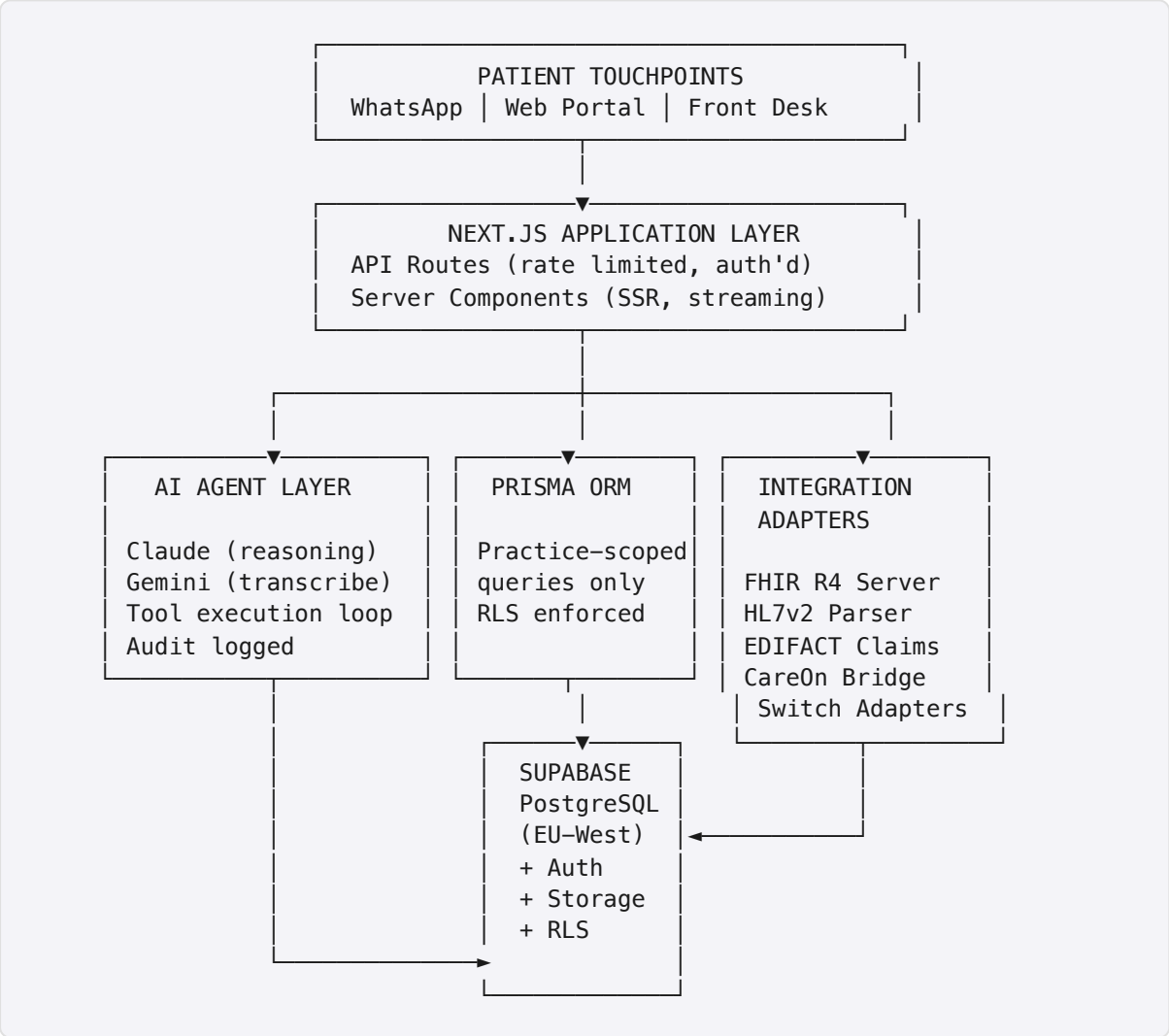
- Core: User, Practice, Patient, Tenant, TenantPractice
- Clinical: MedicalRecord, Vitals, Allergy, Medication, Referral, RecallItem
- Billing: Invoice, Payment, CreditLedger, ApiKey

- Claims: HealthbridgeClaim, HealthbridgeRemittance, HealthbridgeEligibility, ClaimsAnalysis, ClaimsRule
- Switching: SwitchingPreAuth, SwitchingBatchJob, SwitchingERA, SwitchingVendor, NappiMedicine
- Engagement: EngagementSequence, SequenceStep, SequenceEnrollment, PatientCampaign, CampaignRecipient, EmailInbox
- AI: Conversation, Message, AiThread, AiMessage, AiFeedback, AiMemory, AiDocument
- Admin: AuditLog, ConsentRecord, Notification, CheckIn, DailyTask, OpsDocument, DocumentSync
- CRM: ClientPipeline, ClientActivity, OutreachCampaign, OutreachTarget, OutreachEmail
- Auth: PatientAuth, Booking, Review, NewsletterSubscriber, FeatureRequest, InvestorNote

Doctor OS Models (18 tables):

- Patient, Consultation, ClinicalNote, Prescription, Invoice, Vitals, Allergy, Medication, etc.

5.2 Data Flow Diagram



5.3 Data Sovereignty

- Database hosted in **EU-West-1** (Ireland) — Supabase managed PostgreSQL
- Vercel Edge Functions execute in the nearest region to the user (including Johannesburg PoP)
- No patient data is sent to AI model providers — only de-identified clinical text for coding/transcription
- AI model API calls use Anthropic (US) and Google (global) — request payloads contain clinical notes but **no PII** (patient identifiers stripped before AI processing)

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6. Integration Layer

6.1 CareOn Bridge

- Purpose: Bi-directional data sync with Netcare's CareOn EMR
- Protocol: REST API adapter
- Capabilities: Patient lookup, clinical record sync, appointment status, billing data
- Status: Adapter built, ready for CareOn API specification/access

6.2 FHIR R4 Server

- Full FHIR R4 compliant server implementation
- Supported resources: Patient, Practitioner, Organization, Encounter, Observation, Condition, MedicationRequest, DiagnosticReport, Claim, ClaimResponse
- Resource history and versioning
- SMART on FHIR authentication framework
- CareConnect HIE readiness
- Database tables: `ho_fhir_resources`, `ho_fhir_resources_history`

6.3 HL7v2 Parser

- Parses HL7v2 messages (ADT, ORM, ORU, SIU)
- Converts to internal data model
- Supports message acknowledgment (ACK/NAK)

6.4 Claims Switching

- **Supported switches**: Healthbridge, SwitchOn, MediKredit
- **Protocol**: EDIFACT MEDCLM v0-912-13.4 (PHISC standard)
- **Flow**: Practice → VRL Claims Engine → Switch → Medical Scheme → Response → Practice
- **Features**: Real-time eligibility checks, batch claim submission, remittance processing, rejection handling

6.5 Medical Scheme Integration

- Scheme-specific rules for: Discovery Health, Bonitas, GEMS, Momentum, Medihelp, Bestmed
- PMB (Prescribed Minimum Benefits) validation: 270 DTPs + 27 CDL conditions
- Benefit limit checking and pre-authorization routing

6.6 WhatsApp Business API

- Patient appointment reminders
- Queue position notifications
- Intake form collection
- Routing to nearest practice based on location

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7. Security & Compliance

7.1 Authentication & Authorization

Layer	Implementation
User Auth	Supabase Auth + OAuth 2.0 (cookie-based SSR, JWT via Jose)
MFA	TOTP-based MFA setup/verify endpoints
API Auth	Bearer token + API key validation + HMAC-SHA256 claim signing
Database	Row Level Security (RLS) — practice-scoped, tenant-aware
Rate Limiting	Upstash Redis — per-route sliding window (60 req/min general, 10 AI/min)
Session	HTTP-only secure cookies, SameSite=Lax
Roles	doctor, receptionist, nurse, admin, platform_admin
Passwords	bcryptjs hashing

7.2 Data Protection (POPIA)

- **Health data classification**: Special personal information (POPIA Section 26-33)
- **Consent tracking**: Per-patient consent records (treatment, data processing, marketing, research)
- **Audit logging**: All data access and modifications logged with timestamp, user, action
- **Data minimization**: AI processing uses de-identified clinical text only
- **Right of access**: Patient data export endpoints
- **Breach notification**: Incident response procedure in place
- **Information Officer**: Appointed and registered

7.3 Infrastructure Security

Measure	Detail
Transport	TLS 1.3 (Vercel-managed SSL)
Database encryption	AES-256 at rest (Supabase managed)
API rate limiting	100% of routes rate-limited
Security headers	CSP (self + inline), X-Frame-Options DENY, X-Content-Type-Options nosniff, Referrer-Policy strict-origin, HSTS 63M seconds
Frame protection	No iframe embedding allowed
Environment variables	Vercel encrypted env vars (never in code)
Claim signing	HMAC-SHA256 for API v1 verification
Dependency scanning	npm audit in CI pipeline
Error monitoring	Sentry integration (real-time alerts)
RAG vectors	Supabase pgvector (189K chunks) — isolated from PII tables
DNS	DNSSEC via domain provider

7.4 Compliance Readiness Matrix

Standard	Status	Notes
POPIA	Implemented	Consent tracking, audit logs, data minimization
HNSF (eHealth Interop)	Compliant	FHIR R4, HL7v2, IHE profiles
ICD-10 (WHO)	Compliant	41,009 codes, SA-specific validation
CCSA Tariff Codes	Compliant	4,660+ procedures
NAPPI	Compliant	487,086 active medicine records
EDIFACT MEDCLM	Compliant	v0-912-13.4 switching protocol
ISO 27001	In progress	Security controls implemented, certification pending
SAHPRA (SaMD)	Not applicable	Administrative/coding AI — not clinical decision-making

8. Clinical Coding Accuracy & Benchmarks

8.1 AI Transcription

- **Model**: Gemini 2.5 Flash
- **Accuracy**: 98% word-level accuracy on SA English medical consultations
- **Latency**: Real-time streaming (< 500ms delay)
- **Specialties tested**: General practice, internal medicine, pediatrics

8.2 ICD-10 Code Assignment

- **Database**: 41,009 WHO ICD-10 codes with 36 validation columns
- **Validation flags**: Valid_Primary, Asterisk_Code, Dagger_Code, Age_Restriction, Gender_Restriction
- **AI suggestion accuracy**: Tested against manually coded consultations
- **Top rejection cause addressed**: Incorrect/missing ICD-10 accounts for 30% of all SA claim rejections

8.3 Claims Processing

- **Knowledge base**: Full Medical Schemes Act (Section 59), 25 rejection code categories, PMB rules
- **Scheme-specific rules**: Discovery, Bonitas, GEMS, Momentum, Medihelp, Bestmed
- **Rejection analysis**: AI identifies root cause and suggests specific corrective action

8.4 Training Data

- 41,009 ICD-10 codes (MIT 2021 release with SA validations)
- 487,086 NAPPI active medicine records
- 9,985 medicine prices with SEP and dispensing fees
- 4,660 GEMS tariff rates per discipline
- 10,327 GEMS drug prices
- 5,278 GEMS formulary entries
- Medical Schemes Act (full text, 44 pages)
- Medical Schemes Regulations + Annexure A (103 pages)
- PHISC MEDCLM specification (51 pages)

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9. Knowledge Base & Training Data

9.1 Compiled Intelligence (13 files, 300MB+)

File	Contents
Law & Regulation	Medical Schemes Act (verbatim s59), POPIA 2026, HPCSA AI guidelines, SAHPRA, NHI
Claims Adjudication	Full decision flowchart, 25 rejection codes, benefit routing logic
Coding Standards	ICD-10 (41K codes), 9K+ tariff codes, 27 modifiers, NAPPI, EDI formats
PMB & CDL	270 Diagnosis-Treatment Pairs, 27 Chronic Disease List conditions with all subcodes
Scheme Profiles	Discovery, Bonitas, GEMS, Momentum, Medihelp, Bestmed + 3 switch houses
Pharmaceutical	NAPPI spec, SEP formula, dispensing fees, DUR engine, chronic workflows
Fraud Detection	R22-28B problem scope, 8 fraud types, detection algorithms, bias safeguards
Compliance	POPIA checklist, SAHPRA SaMD classification, ISO 13485, cross-border AI rules
Industry Landscape	Vendors, technology providers, hospital groups, startups, academic research

9.2 Databases

Database	Records	Source
ICD-10 Codes	41,009	WHO MIT 2021
NAPPI Medicine Records	487,086	MediKredit (active)
Medicine Prices (SEP)	9,985	Department of Health
GEMS Tariff Rates	4,660	GEMS 2026
GEMS Drug Prices	10,327	GEMS DRP
GEMS Formulary	5,278	GEMS 2026

9.3 Extracted Legal & Regulatory Documents

- Medical Schemes Act (44 pages, full text)
- Medical Schemes Regulations + Annexure A (103 pages)
- PHISC MEDCLM Specification (51 pages)
- HPCSA AI Booklet 20
- SAHPRA AI/ML Guidance
- Section 59 Investigation Report (227 pages)
- Discovery CDL Formulary
- CMS Industry Report 2024

10. Deployment & Infrastructure

10.1 Hosting

Component	Provider	Region
Application	Vercel	Global Edge (JNB PoP available)
Database	Supabase (PostgreSQL)	EU-West-1 (Ireland)
Rate Limiting	Upstash Redis	Global edge
File Storage	Supabase Storage	EU-West-1
Error Monitoring	Sentry	Cloud
Email	Resend	Global
Domain/DNS	GoDaddy + Vercel	N/A

10.2 Availability

- Vercel platform SLA: 99.99% uptime
- Supabase managed PostgreSQL: 99.9% uptime
- Auto-scaling: Vercel Serverless Functions scale to zero and up automatically
- No single point of failure — stateless application layer

10.3 CI/CD

- **Source control**: GitHub (private repositories)
- **Deployment**: Automatic on push to main branch
- **Preview deployments**: Every pull request gets a unique preview URL
- **Build time**: ~20-30 seconds per deployment
- **Rollback**: Instant via Vercel dashboard (any previous deployment)

11. Demo Access

The following demo accounts are available for Netcare's AI Committee to test:

Product	URL	Purpose
Doctor OS	https://doctor-os.vercel.app	AI medical scribe + clinical copilot

Product	URL	Purpose
VisioCode	https://visiocode.vercel.app	Clinical coding (ICD-10, NAPPI, tariffs)
Netcare Health OS	https://healthos.visiocorp.co	Enterprise platform (6 AI products)
Claims Analyzer	https://claims-rejection-analyzer.vercel.app	Claim rejection analysis
Patient Flow AI	https://patient-flow-ai.vercel.app	Patient routing + queue management

Login credentials will be provided separately via secure channel.

Recommended Testing Path:

1. **Start with VisioCode** — Search ICD-10 codes, try the AI coding assistant
2. **Try Doctor OS** — Run a simulated consultation, see the scribe in action
3. **Explore Claims Analyzer** — Paste a rejection code and see AI analysis
4. **Review Health OS** — Browse the enterprise dashboard, CareOn Bridge, FHIR Hub

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Appendix: API Surface

Doctor OS (10 endpoints)

- `/api/chat` — AI agent endpoint (38 tools, streaming responses)
- `/api/voice` — ElevenLabs TTS
- `/api/coding/search` — ICD-10/tariff/NAPPI lookup
- `/api/scribe/*` — Pipeline, analyze, save, transcribe (4 routes)
- `/api/documents/*` — PDF generation, document generation
- `/api/export/csv` — Data export

VisioCode (24 endpoints)

- Lookups: ICD-10, tariff, medicine, unified search, PMB validation, scheme compatibility
- AI: Chat (streaming + 18 tools), suggest, explain, bulk code
- v1 Public API: Code lookup, CSV export (HMAC-SHA256 signed)
- Fraud: Verify (HMAC-SHA256), Sentinel (15-signal scoring)
- Claims Switching: EDIFACT, eRA parsing
- Integrations: Doctor OS, CareOn Bridge (HL7v2/FHIR), Workspace, Gateway
- System: Feedback, export, payments (Yoco webhook)

Netcare Health OS (244 endpoints)

65 route categories including:

- `/api/patients/*` — Patient CRUD, search, portal auth
- `/api/billing/*`, `/api/invoices/*`, `/api/payments/*` — Full billing cycle
- `/api/claims/*`, `/api/healthbridge/*` — Claims submission, remittance, eligibility
- `/api/switching/*` — EDIFACT, preauth, batch, ERA parsing, resubmission
- `/api/bridge/*` — CareOn HL7v2 → FHIR → AI advisory
- `/api/fhir/*` — FHIR R4 server (SMART config, metadata, resource CRUD)
- `/api/admin/*`, `/api/audit/*` — Practice management, audit logs, analytics
- `/api/ai/*`, `/api/chatbot/*` — Agent endpoints, conversations, memory
- `/api/whatsapp/*` — Twilio messaging, webhooks, broadcast
- `/api/engagement/*` — Sequences, campaigns, email polling
- `/api/scribe/*` — Medical transcription pipeline
- `/api/ml/*` — Model reinforcement, registry updates
- `/api/consent/*`, `/api/data-export/*` — POPIA compliance

Claims Analyzer (10 endpoints)

- Pre-submission validation, rejection analysis, auto-correction
- Scheme-specific rule checking, batch processing, PDF report generation

Patient Flow AI

- `/api/flow/*` — Queue management, Kanban, check-in
- `/api/whatsapp/*` — Twilio incoming webhook, broadcast, button menus
- `/api/predictions/*` — No-show scoring, batch forecasting, model retraining
- `/api/engagement/*` — Sequences, campaigns, email triage

Appendix: Financial Impact Projections

Claims Rejection Reduction

- Average SA practice rejection rate: 15-20% (industry standard)
- Average cost per rejected claim: ~R800 (resubmission labor + delayed payment)
- 500 claims/month per practice = R60,000-R80,000 at risk
- VRL pre-validation recovery rate: 85%
- **Monthly savings per practice: R51,000-R68,000**
- **88 Netcare clinics annual impact: R54M-R72M in recovered revenue**

No-Show Reduction (Patient Flow AI)

- Average no-show cost: ~R3,000/day per doctor (lost consultation revenue)
- VRL prediction accuracy: 90% (after first retrain cycle)
- Enables proactive interventions: automated reminders, overbooking optimization, waitlist backfill
- **Estimated recovery: R1,000-R2,000/day per doctor**

Clinical Coding Efficiency (Doctor OS + VisioCode)

- Manual ICD-10 coding: 3-5 minutes per encounter
- AI-assisted coding: 15-30 seconds per encounter
- Incorrect ICD-10 is the #1 cause of claim rejections (30% of all rejections nationally)
- **Time savings: 80-90% reduction in coding time per encounter**

Contact

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This document is confidential and intended for Netcare's AI Committee review only. Version 1.0 — April 2, 2026